

# DELL POWERMAX

## Dell PowerMax 2500 and 8500

The latest [PowerMax models](#) redefine performance at scale, delivering industry-leading cyber resiliency, AI-driven automation, and exceptional efficiency to help you unlock the full potential of your data. Built on the innovative PowerMaxOS 10 and a cutting-edge NVMe scale-out architecture, these systems incorporate advanced global inline data reduction to provide unmatched adaptability, high availability, and performance—empowering your organization to stay ahead in a competitive landscape.

The PowerMax 2500 offers a compelling solution for mission-critical storage, delivering up to 7x<sup>1</sup> more capacity (8 PBe) in half the footprint of previous models. With the industry's most comprehensive data services, the 2500 is designed to handle demanding mixed workloads—block, file, and mainframe—while ensuring the highest levels of data availability and [cyber resiliency](#). Both the PowerMax 2500 and 8500 models come with an [industry-leading data reduction](#) guarantee of 5:1 for Open Systems and 3:1 for Mainframe, providing exceptional efficiency.

For organizations requiring massive consolidation, the PowerMax 8500 delivers unparalleled performance at scale. Supporting up to 16 nodes and a total capacity of 18 PBe, it simplifies operations and reduces [total cost of ownership](#) (TCO). Ideal for the most demanding mixed workloads, the 8500 ensures uninterrupted operations and features advanced cyber resiliency, including Cyber Recovery Service for PowerMax (cyber vault), offered through Dell Professional Services.

Powered by the [Dynamic Fabric architecture](#) and Flexible RAID, PowerMax offers a scalable, flexible design that allows independent growth of nodes and storage capacity in single-drive increments. Leveraging Intel® Xeon® Scalable processors and the latest storage technologies—end-to-end NVMe, InfiniBand 100Gb/s, dual-ported NVMe flash drives, NVMe/TCP connectivity, and hardware-based data reduction—PowerMax is designed for 6 nines of availability. Additionally, it features intelligent PDUs for real-time power monitoring and alerting, ensuring continuously modern storage through [Dell's Future-Proof program](#).

## Specifications

### Scale up and Scale out

PowerMax is built from modular storage components for compute and storage media. The compute modules are packaged as node pairs. Each node pair contains two PowerMax compute nodes, complete software and licensing, cache memory, redundant power, and connectivity modules. These are combined with 48-slot Dynamic Media Enclosures (DMEs) to configure NVMe flash drives. PowerMax arrays are delivered with the Inclusive Software package. NVMe drive capacity can be added to the system to scale up to a total effective capacity of 8 PBe on the PowerMax 2500 and up to 18 PBe on the PowerMax 8500.

Detailed specifications and a comparison of the PowerMax 2500 and 8500 arrays follow:

<sup>1</sup>Based on Dell's internal analysis comparing Effective Storage Capacity of the PowerMax 2500 compared with the PowerMax 2000, April 2025. Actual storage capacities will vary.

| Array family                                  | PowerMax 2500                                                                                                                                                                                                                                                              | PowerMax 8500                                                                                                                                                                                                                                                              |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Node Pairs</b>                             |                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                            |
| NUMBER OF NODE PAIRS                          | 1 to 2                                                                                                                                                                                                                                                                     | 1 to 8                                                                                                                                                                                                                                                                     |
| NODE PAIR MODULE                              | 3U                                                                                                                                                                                                                                                                         | 3U                                                                                                                                                                                                                                                                         |
| CPU                                           | Memory config 1 <sup>6</sup> -3: Intel Xeon Gold 5218 2.8 GHz with 16 core <sup>1</sup>                                                                                                                                                                                    | Memory config 2-3: Intel Xeon Gold 6254 3.9 GHz with 18 core <sup>1</sup>                                                                                                                                                                                                  |
|                                               | Memory config 4 and 5: Intel Xeon Gold 6240L                                                                                                                                                                                                                               | Memory Config 4 and 5: Intel Xeon Gold 8280L                                                                                                                                                                                                                               |
| CORE NUMBER PER CPU/PER NODE PAIR/PER SYSTEM  | Memcfg 1 <sup>6</sup> -3: 16/64/128<br>Memcfg 4 and 5: 18/72/144 <sup>4</sup>                                                                                                                                                                                              | Memcfg 1-3: 18/72/576<br>Memcfg 4 and 5: 20/112/736 <sup>3,4</sup>                                                                                                                                                                                                         |
| DYNAMIC FABRIC                                | Direct Connection InfiniBand:<br>100 Gbps per port                                                                                                                                                                                                                         | InfiniBand Dual Redundant Fabric:<br>100 Gbps per port                                                                                                                                                                                                                     |
| <b>CACHE</b>                                  |                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                            |
| CACHE-SYSTEM MIN (RAW)                        | 896GB                                                                                                                                                                                                                                                                      | 1792GB                                                                                                                                                                                                                                                                     |
| CACHE-SYSTEM MAX (RAW)                        | 15.36TB                                                                                                                                                                                                                                                                    | 45.056TB <sup>3</sup>                                                                                                                                                                                                                                                      |
| CACHE-PER NODE PAIR OPTIONS                   | 896GB, 1.792TB, 3.584TB, 7.680TB                                                                                                                                                                                                                                           | 1.792TB, 3.584TB, 7.680TB                                                                                                                                                                                                                                                  |
| <b>VAULT</b>                                  |                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                            |
| VAULT STRATEGY                                | Vault to Flash                                                                                                                                                                                                                                                             | Vault to Flash                                                                                                                                                                                                                                                             |
| VAULT IMPLEMENTATION                          | 2 to 4 NVMe SED Flash Module/Node Pair <sup>2</sup>                                                                                                                                                                                                                        | 4 NVMe SED Flash Module /Node Pair <sup>2</sup>                                                                                                                                                                                                                            |
| <b>FRONT-END I/O MODULES</b>                  |                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                            |
| MAX. FRONT-END I/O MODULES/NODE PAIR          | 8                                                                                                                                                                                                                                                                          | 8                                                                                                                                                                                                                                                                          |
| FRONT-END I/O MODULES AND PROTOCOLS SUPPORTED | 2 x 100Gb/s (Ethernet, iSCSI, SRDF, NVMe/TCP)<br>4 x 64Gb/s (FC, NVMe, SRDF) <sup>5</sup><br>4 x 32Gb/s (FC, NVMe/FC, FICON, SRDF)<br>4 x 25Gb/s (Ethernet, iSCSI, SRDF, NVMe/TCP)<br>4 x 10Gb/s (Ethernet, iSCSI, SRDF, NVMe/TCP)<br>1 x zHyperlink Port (MF, zHyperlink) | 2 x 100Gb/s (Ethernet, iSCSI, SRDF, NVMe/TCP)<br>4 x 64Gb/s (FC, NVMe, SRDF) <sup>5</sup><br>4 x 32Gb/s (FC, NVMe/FC, FICON, SRDF)<br>4 x 25Gb/s (Ethernet, iSCSI, SRDF, NVMe/TCP)<br>4 x 10Gb/s (Ethernet, iSCSI, SRDF, NVMe/TCP)<br>1 x zHyperlink Port (MF, zHyperlink) |
| <b>POWERMAX FILE MODULES</b>                  |                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                            |
| MAX FILE I/O MODULES/SOFTWARE NODES           | 4                                                                                                                                                                                                                                                                          | 4                                                                                                                                                                                                                                                                          |
| FILE I/O MODULES SUPPORTED                    | 10 Gb/s: 4 x 10Gb/s File<br>25 Gb/s: 4 x 25Gb/s File                                                                                                                                                                                                                       | 10 Gb/s: 4 x 10Gb/s File<br>25 Gb/s: 4 x 25Gb/s File                                                                                                                                                                                                                       |
| <b>POWERMAX FILE SOFTWARE NODES</b>           |                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                            |
| MAX SOFTWARE FILE NODES                       | 4 (1 per Node, 2 per Node pair)                                                                                                                                                                                                                                            | 8 (1 per Node, 2 per Node pair)                                                                                                                                                                                                                                            |
| MAX FILE CAPACITY/ARRAY (PETABYTES USABLE)    | 8PiBe                                                                                                                                                                                                                                                                      | 18PiBe                                                                                                                                                                                                                                                                     |

<sup>1</sup> CPUs run continuously in turbo mode, except at significantly high ambient temperatures.

<sup>2</sup> Encryption will be disabled if not ordered.

<sup>3</sup> Memory Config 4 is limited to a maximum of 4 node pairs in PowerMax 8500.

<sup>4</sup> Expanded cores only for Memory Config 4 and 5.

<sup>5</sup> Only multi-mode support is available.

<sup>6</sup> Not available with SRDF in PowerMax 2500.

| Array family                                       | PowerMax 2500                                                                            |                                              | PowerMax 8500                                                 |                                              |
|----------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------|----------------------------------------------|
| CAPACITY, DRIVES                                   |                                                                                          |                                              |                                                               |                                              |
| Max Capacity per Array (Open) <sup>1,7</sup>       | 8PiBe / 8.8 PBe                                                                          |                                              | 18 PiBe / 20 PBe                                              |                                              |
| Base capacity (Open)                               | 30.72TBu                                                                                 |                                              | 30.72Tbu                                                      |                                              |
| Max Capacity per Array (Mainframe) <sup>7, 8</sup> | 3.8PiBe / 4.1PBe                                                                         |                                              | 9.8 PiBe / 10.7PBe                                            |                                              |
| Base capacity (Mainframe)                          | 15.36TBu                                                                                 |                                              | 15.36Tbu                                                      |                                              |
| Incremental Flash Capacity Upgrades                | TLC: 3.84TB, 7.68TB, 15.36TB, 30.72TB <sup>3</sup><br>QLC: 15.36TB, 30.72TB <sup>3</sup> |                                              | TLC: 3.84TB, 7.68TB, 15.36TB, 30.72TB <sup>3</sup>            |                                              |
| Maximum Drives per Array                           | 96                                                                                       |                                              | 384                                                           |                                              |
| Maximum Drives per System Bay                      | 96/192/288 <sup>2</sup>                                                                  |                                              | 192/384                                                       |                                              |
| Minimum Drive Count per System                     | 6 (Mainframe) / 10 (Open)                                                                |                                              | 6 (Mainframe) / 10 (Open)                                     |                                              |
| NVMe DRIVES                                        |                                                                                          |                                              |                                                               |                                              |
| NVMe units accepted (2.5 in.)                      | 3.84TB, 7.68TB, 15.36TB <sup>9</sup> , 30.72TB <sup>3,9</sup>                            |                                              | 3.84TB, 7.68TB, 15.36TB <sup>9</sup> , 30.72TB <sup>3,9</sup> |                                              |
| Interface BE                                       | Dual PCIe Gen 3 x8 NVMe attached interface                                               |                                              | NVMe/NVMeoF via Dual 100G Infiniband Fabric                   |                                              |
| Flexible RAID options with support                 | RAID 1 (1+1)<br>RAID 5 (4+1) <sup>6</sup><br>RAID 5 (8+1)<br>RAID 5 (12+1)               | RAID 6 (12+2)<br>RAID 6 (24+2) <sup>11</sup> | RAID 1 (1+1)<br>RAID 5 (8+1)<br>RAID 5 (12+1)                 | RAID 6 (12+2)<br>RAID 6 (24+2) <sup>11</sup> |
| Mixed RAID group support                           | No                                                                                       |                                              | No                                                            |                                              |
| Support for Mixed Drive Capacities                 | Yes <sup>3</sup>                                                                         |                                              | Yes <sup>3</sup>                                              |                                              |
| NVMe DYNAMIC MEDIA ENCLOSURE                       |                                                                                          |                                              |                                                               |                                              |
| 48 x 2.5" Drive DME                                | Yes                                                                                      |                                              | Yes                                                           |                                              |
| CABINET SETTINGS                                   |                                                                                          |                                              |                                                               |                                              |
| Standard 19" bays                                  | Yes                                                                                      |                                              | Yes                                                           |                                              |
| System Bay configurations                          | Up to 3 Systems/Bay                                                                      |                                              | Up to 6 Node Pairs/Bay <sup>4,5</sup>                         |                                              |
| Third-party rack mount option                      | Yes                                                                                      |                                              | Yes                                                           |                                              |
| DISPERSION                                         |                                                                                          |                                              |                                                               |                                              |
| Standard and third-party enclosures                | N/A — single floor tile system                                                           |                                              | Yes                                                           |                                              |
| PRE-CONFIGURATION FROM FACTORY                     |                                                                                          |                                              |                                                               |                                              |
| 100% Thin Provisioned                              | Yes                                                                                      |                                              | Yes                                                           |                                              |
| HOST SUPPORT                                       |                                                                                          |                                              |                                                               |                                              |
| Open Systems                                       | Yes                                                                                      |                                              | Yes                                                           |                                              |
| Mainframe                                          | Yes                                                                                      |                                              | Yes                                                           |                                              |
| Mixed Mainframe and Open Systems                   | Yes                                                                                      |                                              | Yes                                                           |                                              |
| POWER OPTIONS                                      |                                                                                          |                                              |                                                               |                                              |
| Input power options                                | Single phase or three-phase<br>Delta or Wye                                              |                                              | Single phase or three-phase<br>Delta or Wye                   |                                              |
| POWER DISTRIBUTION UNIT                            |                                                                                          |                                              |                                                               |                                              |
| Intelligent PDU                                    | Default <sup>10</sup>                                                                    |                                              | Default <sup>10</sup>                                         |                                              |

<sup>1</sup> Maximum capacity per array based on 5:1 Data Reduction.

<sup>2</sup> 288 drives can be supported in a single cabinet when three systems are packaged in the same rack.

<sup>3</sup> Up to two consecutive compatible drive capacities, e.g. 3.84TB and 7.68TB are supported per storage resource pool (SRP.)

<sup>4</sup> This is based on a dense configuration. System bay configuration can also support balanced configuration.

<sup>5</sup> Dense configurations allow 6 node pairs in System Bay 1 and 2 additional node pairs in System Bay 2.

<sup>6</sup> R5(4+1) is MF only and supports only 3.84TB drives.

<sup>7</sup> PB is base-10 decimal notation (1000x1000x1000x1000x1000). PiB is base-2 binary notation (1024x1024x1024x1024x1024).

<sup>8</sup> Maximum mainframe capacity is based on 3:1 Data Reduction.

<sup>9</sup> Refer to Table 1 (page 4) for more information.

<sup>10</sup> PowerMax offers an intelligent PDU by default with PowerMax 10.1 release that enables real-time telemetry and monitoring of power, voltage, current, external temperature, and humidity.

<sup>11</sup> Only 15.36TB and 30.72TB drive capacities are supported with R6(24+2).

| Array family                                                                                              | PowerMax 2500 | PowerMax 8500 |
|-----------------------------------------------------------------------------------------------------------|---------------|---------------|
| <b>SRDF AND FRONT-END I/O PROTOCOLS COMPATIBLE</b>                                                        |               |               |
| <b>100 Gb/s Ethernet Host Ports, iSCSI Ports, NVMe/TCP Ports, SRDF Ports</b>                              |               |               |
| Maximum/Node Pair                                                                                         | 16            | 16            |
| Maximum number per array                                                                                  | 32            | 128           |
| <b>64 Gb/s FC Host Ports, NVMe Ports, SRDF Ports</b>                                                      |               |               |
| Maximum/Node Pair                                                                                         | 32            | 32            |
| Maximum number per array                                                                                  | 64            | 256           |
| <b>32 Gb/s FC Host Ports, FICON Ports, SRDF Ports</b>                                                     |               |               |
| Maximum/Node Pair                                                                                         | 32            | 32            |
| Maximum number per array                                                                                  | 64            | 256           |
| <b>25 Gb/s Ethernet Host Ports, iSCSI Ports (Optical), SRDF Ports (Optical), NVMe/TCP Ports (Optical)</b> |               |               |
| Maximum/Node Pair                                                                                         | 32            | 32            |
| Maximum number per array                                                                                  | 64            | 256           |
| <b>10 Gb/s Ethernet Host Ports, iSCSI Ports (Optical), SRDF Ports (Optical), NVMe/TCP Ports (Optical)</b> |               |               |
| Maximum/Node Pair                                                                                         | 32            | 32            |
| Maximum number per array                                                                                  | 64            | 256           |
| <b>zHyperlink ports</b>                                                                                   |               |               |
| Maximum number of ports/Node Pair                                                                         | 2             | 2             |
| Maximum number of ports per array                                                                         | 4             | 4             |
| <b>EMBEDDED File Ports</b>                                                                                |               |               |
| <b>10 Gb/s File ports</b>                                                                                 |               |               |
| Maximum number of software data transfer/ports                                                            | 16            | 16            |
| Maximum number of ports per array                                                                         | 64            | 256           |
| <b>25 Gb/s File ports</b>                                                                                 |               |               |
| Maximum number of software data transfer/ports                                                            | 16            | 16            |
| Maximum number of ports per array                                                                         | 64            | 256           |

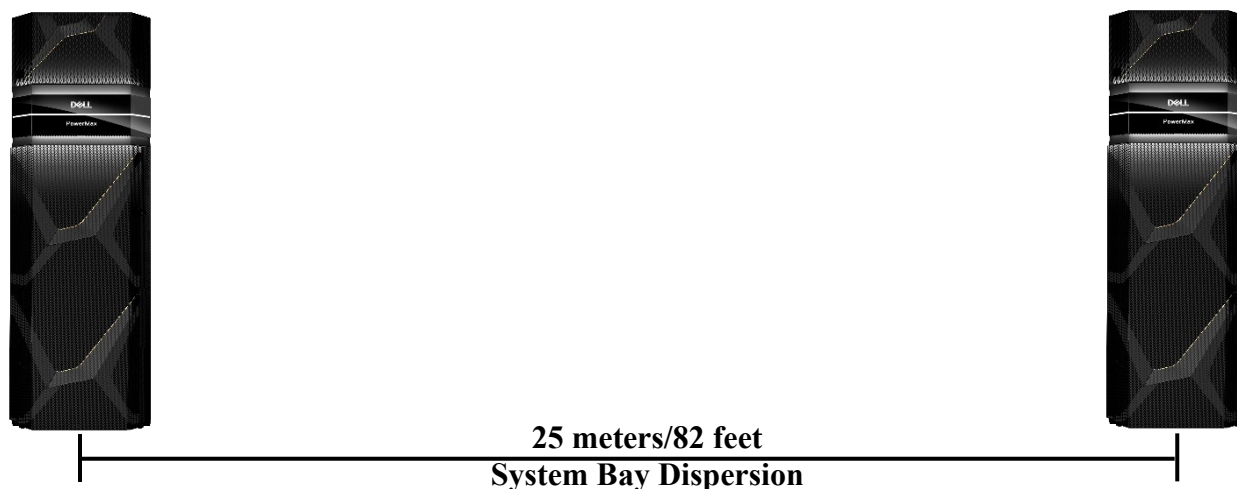
Table 1 – QLC/TLC 15 TB and 30 TB drives RAID support

| Drive     | R1(1+1) | R5(4+1) | R5(8+1)   | R5(12+1) | R6(12+2)               | R6(24+2) |
|-----------|---------|---------|-----------|----------|------------------------|----------|
| QLC 15.36 | No      | No      | 2500 Only | No       | 2500 Only              | No       |
| QLC 30.72 | No      | No      | No        | No       | 2500 Only <sup>1</sup> | No       |
| TLC 15.36 | Yes     | No      | Yes       | Yes      | Yes                    | Yes      |
| TLC 30.72 | No      | No      | Yes       | No       | Yes                    | Yes      |

<sup>1</sup>Supported with Mem Config 2, 3, 4, 5 (not 1)

## Figure 1 - System bay dispersion

System Bay Dispersion allows customers to separate any individual or contiguous group of system bays by up to a distance of 82 feet (25 meters) from System Bay 1. This provides unsurpassed datacenter flexibility in solving floor loading constraints or working around obstacles that might preclude fully contiguous configurations. This is applicable only to the PowerMax 8500, as the PowerMax 2500 is a single-bay solution.



## Support for flash drives

The PowerMax 2500 and PowerMax 8500 support the latest dual-ported native NVMe Flash units. All drives support two independent I/O channels with automatic failover fault isolation. Consult your Dell sales representative for the latest list of supported drives and types. All capacities are based on 1 GB = 1,000,000,000 bytes. Actual usable capacity may vary by configuration.

Table 2 - 2.5" NVMe flash drives used in base systems and capacity pack upgrades

| Platform support                                       | PowerMax 2500/8500 | PowerMax 2500/8500 | PowerMax 2500/8500 | PowerMax 2500/8500<br>Mem Config 4 Only |
|--------------------------------------------------------|--------------------|--------------------|--------------------|-----------------------------------------|
| Nominal capacity (GB)                                  | 3840 <sup>1</sup>  | 7680 <sup>1</sup>  | 15360 <sup>1</sup> | 30720 <sup>1</sup>                      |
| Type                                                   | NVMe Flash         | NVMe Flash         | NVMe Flash         | NVMe Flash                              |
| Raw capacity (GB)                                      | 3840               | 7680               | 15360              | 30720                                   |
| Open systems<br>formatted capacity (GB) <sup>2</sup>   | 3840.30            | 7680.61            | 15047.65           | 30095.05                                |
| Mainframe 3390<br>formatted capacity (GB) <sup>2</sup> | 3840.30            | 7680.61            | 15047.65           | 30095.05                                |

<sup>1</sup>In any configuration, capacity upgrades can contain a maximum of two different underlying drive sizes to achieve the best useful capacity desired. This is automatically optimized by the configuration tools.

<sup>2</sup>Formatted capacities shown are for RAID 5 (12+1). Values vary slightly with different RAID types.

**Table 3 - Energy consumption and heat dissipation at <26°C and >35°C**

| Component                                                                                      | PowerMax 2500                         |        |                                  |        | PowerMax 8500                         |        |                                  |        |
|------------------------------------------------------------------------------------------------|---------------------------------------|--------|----------------------------------|--------|---------------------------------------|--------|----------------------------------|--------|
| Maximum power and heat dissipation at temperatures < 26°C <sup>2</sup> and > 35°C <sup>3</sup> | Maximum total power consumption (kVA) |        | Maximum heat dissipation (Btu/h) |        | Maximum total power consumption (kVA) |        | Maximum heat dissipation (Btu/h) |        |
|                                                                                                | < 26°C                                | > 35°C | < 26°C                           | > 35°C | < 26°C                                | > 35°C | < 26°C                           | > 35°C |
| System Cabinet 1, Single (Node Pair, Single DME)<br>PowerMax 2500                              | 2.213                                 | 3.131  | 7,551                            | 10,683 | N/A                                   | N/A    | N/A                              | N/A    |
| System Cabinet 1, Two (Single Node Pair, Single DME)<br>PowerMax 2500                          | 4.426                                 | 6.262  | 15,102                           | 21,366 | N/A                                   | N/A    | N/A                              | N/A    |
| System Cabinet 1, Three (Single Node Pair, Single DME)<br>PowerMax 2500                        | 6.639                                 | 9.393  | 22,654                           | 32,049 | N/A                                   | N/A    | N/A                              | N/A    |
| System Cabinet 1, One (Dual Node Pair, Single DME)<br>PowerMax 2500                            | 3.724                                 | 5.113  | 12,706                           | 17,445 | N/A                                   | N/A    | N/A                              | N/A    |
| System Cabinet 1, Two (Dual Node Pair, Single DME)<br>PowerMax 2500                            | 7.448                                 | 10.225 | 25,412                           | 34,890 | N/A                                   | N/A    | N/A                              | N/A    |
| System Cabinet 1, Three (Dual Node Pair, Single DME)<br>PowerMax 2500                          | 11.171                                | 15.338 | 38,119                           | 52,335 | N/A                                   | N/A    | N/A                              | N/A    |
| System Cabinet 1, One (Dual Node Pair, Dual DME)<br>PowerMax 2500                              | 4.426                                 | 6.262  | 15,102                           | 21,366 | N/A                                   | N/A    | N/A                              | N/A    |
| System Cabinet 1, Two (Dual Node Pair, Dual DME)<br>PowerMax 2500                              | 8.852                                 | 12.524 | 30,205                           | 42,732 | N/A                                   | N/A    | N/A                              | N/A    |
| System Cabinet 1, Three (Dual Node Pair, Dual DME)<br>PowerMax 2500                            | 13.278                                | 18.785 | 45,307                           | 64,099 | N/A                                   | N/A    | N/A                              | N/A    |
| System Cabinet 1, Balanced (Four Node Pair, Four DME)<br>PowerMax 8500                         | N/A                                   | N/A    | N/A                              | N/A    | 11.178                                | 14.736 | 38,140                           | 50,281 |
| System Cabinet 2, Balanced (Four Node Pair, Four DME)<br>PowerMax 8500                         | N/A                                   | N/A    | N/A                              | N/A    | 10.846                                | 14.404 | 37,007                           | 49,148 |
| System Cabinet 1, Dense (Six Node Pair, Four DME)<br>PowerMax 8500                             | N/A                                   | N/A    | N/A                              | N/A    | 14.899                                | 19.376 | 50,839                           | 66,115 |
| System Cabinet 2, Balanced (Two Node Pair, Four DME)<br>PowerMax 8500                          | N/A                                   | N/A    | N/A                              | N/A    | 7.124                                 | 9.764  | 24,308                           | 33,315 |

<sup>1</sup> Power values for configurations with two, three, and four node pairs, placed in the System 2 Enclosure (PowerMax 8500 only)

<sup>2</sup> Values at <26 °C reflect the maximum values in a more stable state during normal operation

<sup>3</sup> Power values and heat dissipations are shown at >35 °C to reflect the higher power levels associated with both the battery recharge cycle and the initiation of high ambient temperature Adaptive Cooling algorithms.

**Table 4 - Physical specifications**

| Component                                                          | Height<br>(in./cm) | Width<br>(in./cm) | Depth<br>(in./cm) | Weight (maximum<br>lb/kg) |
|--------------------------------------------------------------------|--------------------|-------------------|-------------------|---------------------------|
| System Bay 1, Four Node Pair, Four DME (Balanced)<br>PowerMax 8500 | 78.4/199.2         | 23.5/60           | 47.3/120          | 1537/697                  |
| System Bay 2, Four Node Pair, Four DME (Balanced)<br>PowerMax 8500 | 78.4/199.2         | 23.5/60           | 47.3/120          | 1410/640                  |
| System Bay 1, Six Node Pair, Four DME (Dense)<br>PowerMax 8500     | 78.4/199.2         | 23.5/60           | 47.3/120          | 1806/819                  |
| System Bay 2, Dual Node Pair, Four DME (Dense)<br>PowerMax 8500    | 78.4/199.2         | 23.5/60           | 47.3/120          | 1136/515                  |
| System Bay 1, Single Node Pair, Single DME<br>PowerMax 2500        | 78.4/199.2         | 23.5/60           | 45.2/114.8        | 675/306                   |
| System Bay 1, Dual Node Pair, Single DME<br>PowerMax 2500          | 78.4/199.2         | 23.5/60           | 45.2/114.8        | 813/369                   |
| System Bay 1, Dual Node Pair, Dual DME<br>PowerMax 2500            | 78.4/199.2         | 23.5/60           | 45.2/114.8        | 900/408                   |
| System Bay 1, Three Node Pair, Three DME<br>PowerMax 2500          | 78.4/199.2         | 23.5/60           | 45.2/114.8        | 1125/510                  |
| System Bay 1, Four Node Pair, Four DME<br>PowerMax 2500            | 78.4/199.2         | 23.5/60           | 45.2/114.8        | 1375/624                  |
| System Bay 1, Six Node Pair, Six DME<br>PowerMax 2500              | 78.4/199.2         | 23.5/60           | 45.2/114.8        | 1838/834                  |

## Input power requirements

**Table 5 - Single phase North American, International and Australian**

| Specification                                                    | North American 3-wire<br>connection<br>(2 L and 1 G) <sup>1</sup>                                         | International and Australian 3-wire<br>connection<br>(1 L, 1 N and 1 G) <sup>1</sup> |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Input nominal voltage                                            | 200 - 240 VAC +/- 10% L – L nom                                                                           | 220 - 240 VAC +/- 10% L – N nom                                                      |
| Frequency                                                        | 50 - 60 Hz                                                                                                | 50 - 60 Hz                                                                           |
| Circuit Breakers                                                 | 30 A                                                                                                      | 30 or 32 A                                                                           |
| Power zones                                                      | Two                                                                                                       | Two                                                                                  |
| PowerMax 2500 minimum input line cord<br>requirements per system | One-Node Pair, one-DME system: One 30 A or 32 A single-phase line cord per power<br>zone for each system. |                                                                                      |
| PowerMax 2500 maximum input line<br>cord requirements per system | Two-Node Pair, two-DME system: Two 30 A or 32 A single-phase line cords per power<br>zone.                |                                                                                      |
| PowerMax 8500 minimum input line cord<br>requirements per system | One-Node Pair, one-DME system: One 30 A or 32 A single-phase line cord per power<br>zone.                 |                                                                                      |
| PowerMax 8500 maximum input line<br>cord requirements per system | Six-Node Pair, four-DME system in one rack: Six 30 A or 32 A single-phase line cords<br>per power zone.   |                                                                                      |

<sup>1</sup>L = line or phase, N = neutral, G = ground

**Table 6 - Three-phase North American, International, Australian**

| Specification                                            | North American (DELTA) 4-wire connection (3 L and 1 G) <sup>1</sup> | International (WYE) 5-wire connection (3 L, 1 N and 1 G) <sup>1</sup> |
|----------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------|
| Input voltage <sup>2</sup>                               | 200 - 240 VAC +/- 10% L – L nom                                     | 220 - 240 VAC +/- 10% L – N nom                                       |
| Frequency                                                | 50 - 60 Hz                                                          | 50 - 60 Hz                                                            |
| Circuit Breakers                                         | 50 A                                                                | 30/32 A                                                               |
| Power zones                                              | Two                                                                 | Two                                                                   |
| Minimum power requirements at customer site              | One 50 A three-phase line cord per power zone.                      | One 30 A or 32 A three-phase line cord per power zone.                |
| Maximum power requirements at customer site <sup>3</sup> | Two 50 A three-phase line cords per power zone.                     | Two 30 A or 32 A three-phase line cords per power zone.               |

<sup>1</sup>L = line or phase, N = neutral, G = ground

<sup>2</sup>An imbalance of AC input currents may exist on the three-phase power source feeding the array, depending on the configuration. The customer's electrician must be alerted to this possible condition to balance the phase-by-phase loading conditions within the customer's data center

<sup>3</sup>A second input AC line cord must be added for each power zone when the total number of Node Pairs and DAEs (combined) in a rack reaches seven.

**Table 7 - Radio frequency interference**

Electro-magnetic fields which include radio frequencies can interfere with the operation of electronic equipment. Dell products have been certified to withstand radio frequency interference in accordance with EN61000-4-3. In Data Centers that employ intentional radiators, such as cell phone repeaters, the maximum ambient RF field strength should not exceed 3 volts/meter.

| Repeater power level (watts) | Recommended minimum distance (feet/meters) |
|------------------------------|--------------------------------------------|
| 1                            | 9.84 FT (3 M)                              |
| 2                            | 13.12 (4 M)                                |
| 5                            | 19.69 FT (6 M)                             |
| 7                            | 22.97 FT (7 M)                             |
| 10                           | 26.25 FT (8 M)                             |
| 12                           | 29.53 FT (9 M)                             |
| 15                           | 32.81 FT (10 M)                            |



Table 8 - Dell Technologies global services

| Dell Technologies World Class Services |                                                                                                                                                                                                                     |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation services                | <ul style="list-style-type: none"><li>• Dell ProDeploy Enterprise Suite</li><li>• Dell Data Migration Services</li><li>• Dell Residency Services</li><li>• Dell Data Sanitization Services for Enterprise</li></ul> |
| Support services                       | <ul style="list-style-type: none"><li>• Dell ProSupport Enterprise Suite</li><li>• Dell Keep Your Hard Drive for Enterprise</li></ul>                                                                               |
| Managed services                       | <ul style="list-style-type: none"><li>• Dell Managed Services for Storage</li></ul>                                                                                                                                 |
| Dell Technologies Consulting Services  | <ul style="list-style-type: none"><li>• Cyber Recovery Services for PowerMax (Cyber Vault)</li><li>• Advisory Services workshops</li></ul>                                                                          |
| Dell Technologies Education Services   | <ul style="list-style-type: none"><li>• PowerMax technical training courses and certifications</li></ul>                                                                                                            |
| Support technology and services        | <ul style="list-style-type: none"><li>• MyService360</li><li>• Secure Remote Services, SupportAssist Enterprise</li></ul>                                                                                           |

DECLARATION OF CONFORMITY

Dell Technologies IT equipment complies with all applicable regulatory requirements for electromagnetic compatibility, product safety, and environmental standards when placed on the market. Detailed regulatory information and compliance verification are available on the Dell standards compliance website. [http://dell.com/regulatory\\_compliance](http://dell.com/regulatory_compliance)

This product has been tested and verified whether it will work within the permitted range of environmental attributes of ashrae's 2-level operating condition class between 10°C and 35°C and within the corresponding relative humidity range.



[Learn more](#) about Dell PowerMax



[Contact a Dell Expert](#)



[See more](#) features



Join the conversation with #POWERMAX